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Next >

10.3.5 The Francis implicit QR Step

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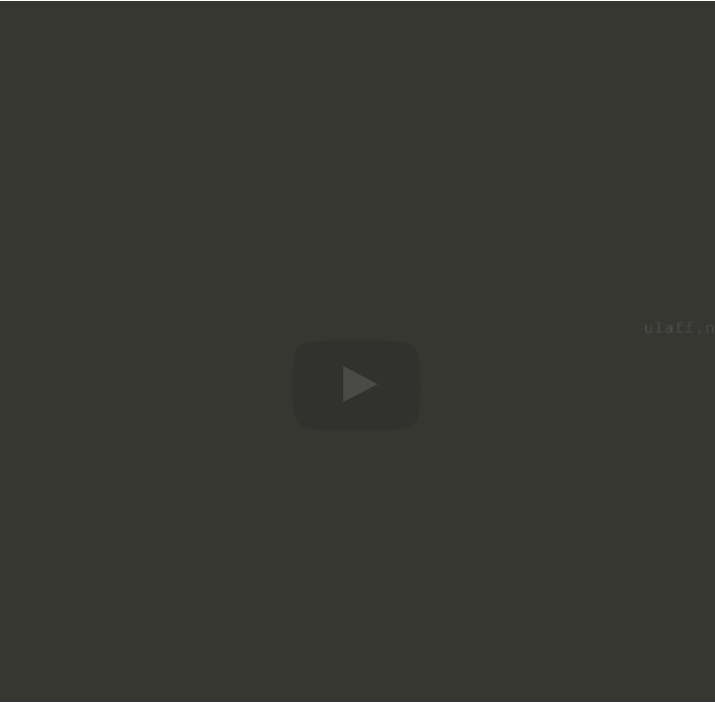
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mean? That means that we don't need to subtract off our shift off of every entry of the diagonal. We merely need to form these new elements right here so we can compute our first Givens rotation. We can take that Givens rotation then back to our original matrix, and apply it from the left and from the right. And now we take this away as well. And we apply this Givens rotation from the left and from the right that introduces a nonzero

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Read Unit 10.3.5 of the notes. [\[LINK\]](#)

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A typical step when "chasing the bulge" one row and column further down the matrix involves the computation

$$\left(\begin{array}{c|cc|c} \alpha_{i-1,i-1} & \times & \times & 0 \\ \hline \hat{\alpha}_{i,i-1} & \hat{\alpha}_{i,i} & \times & 0 \\ 0 & \hat{\alpha}_{i+1,i} & \hat{\alpha}_{i+1,i+1} & \times \\ \hline 0 & \hat{\alpha}_{i+2,i} & \hat{\alpha}_{i+2,i+1} & \alpha_{i+2,i+2} \end{array}\right) = \left(\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & \gamma_i & \sigma_i & 0 \\ 0 & -\sigma_i & \gamma_i & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right) \left(\begin{array}{c|cc|c} \alpha_{i-1,i-1} & \times & \times & 0 \\ \hline \alpha_{i,i-1} & \alpha_{i,i} & \times & 0 \\ \alpha_{i+1,i-1} & \alpha_{i+1,i} & \alpha_{i+1,i+1} & \times \\ \hline 0 & 0 & \alpha_{i+2,i+1} & \alpha_{i+2,i+2} \end{array}\right)$$

Give a strategy (or formula) for computing

$$\left(\begin{array}{c|cc|c} & & & \\ \hline & \hat{\alpha}_{i,i-1} & \hat{\alpha}_{i,i} & \\ & & \hat{\alpha}_{i+1,i} & \hat{\alpha}_{i+1,i+1} \\ \hline & & \hat{\alpha}_{i+2,i} & \hat{\alpha}_{i+2,i+1} \end{array}\right)$$

☒ done



Solution:

For a complete solution, see [the notes](#).

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